

From Patchwork to Progress: Standardizing the Human–AI Collaboration Frontier

Most AI policy discussions focus on what AI systems can do, rather than what people actually do with AI systems in practice. As AI moves from experimental tools to institutional infrastructure, this distinction becomes critical for governance.

INTRODUCTION

Governments, hospitals, schools, and firms worldwide are rapidly deploying artificial intelligence (AI) technology, a shift increasingly formalized through institutional policies and guidance. These deployments are often justified by policy language emphasizing innovation, productivity, efficiency, and “maximizing benefits” from AI. Governments worldwide have incorporated these values into national strategies: over 70 countries now have AI policies that coordinate research, workforce initiatives, and adoption priorities, and intergovernmental principles like the OECD AI Principles are used to benchmark and guide national plans [1].

Policymakers and institutional leaders routinely draw on empirical studies of human–AI collaboration to guide deployment decisions and signal that AI adoption is supported by research rather than speculation [2, 3, 4]. As AI deployment moves from experimental pilots to everyday workflows, studies on human-AI collaboration have expanded at an unprecedented rate. Yet, beneath the surface of these individually rigorous studies lies a persistent “evidence gap” when the literature is viewed as a whole. While landmark field experiments report substantial productivity surges [5, 6, 7], others observe negligible, inconsistent, or even negative effects on professional outcomes [8, 9]. At first glance, these findings appear to suggest that AI’s impact on human outcomes is unpredictable. We argue instead that these discrepancies largely arise from subtle, undocumented differences in how humans and AI systems interact.

In the United States, policy frameworks such as the NIST AI Risk Management Framework calls for “meaningful human oversight” in AI deployment [10], but the framework leaves open how oversight should be empirically evaluated in practice. This disconnect exists because oversight is currently treated as a static legal requirement rather than a variable experimental condition, leaving the specific mechanisms—such as real-time assistance versus post-hoc verification—undocumented. Consequently, scientific reporting standards have not kept pace with the policy mandates they are intended to support, with human–AI interaction often treated as an implicit variable in both research and policy discourse. This paper outlines a tangible solution that draws on multidisciplinary precedents to document how people and AI systems interact in practice.

THE STATE OF HUMAN-AI COLLABORATION EVIDENCE

Human–AI collaboration research now spans many domains, including business, healthcare, education, and public administration. Within this literature, randomized controlled trials (RCTs) are often treated as especially policy-relevant because they appear to offer clear, causal answers to practical questions like: does AI assistance improve productivity or accuracy

compared to existing baselines? In domains such as healthcare, economics, and education, RCT evidence is frequently treated as the gold standard for measuring the effectiveness of a new intervention when justifying large-scale deployment decisions. Yet, the design of any RCT fundamentally shapes the causal effect being estimated. This sensitivity is especially serious in the rapidly evolving field of human–AI collaboration, where technologies change quickly and studies vary widely in their experimental design, participant populations, AI systems, and the tasks used to measure outcomes.

In human–AI collaboration studies, seemingly minor design choices can substantially alter the causal effect being measured. The same AI system may improve outcomes for novice users while degrading performance for experts, depending on how recommendations are presented and whether users are trained to interpret them. Differences in interface design—such as real-time suggestions versus post hoc review—can shape reliance, effort, and error correction in ways that affect measured performance. Even when tasks appear similar across studies, variations in evaluation criteria (for example, speed versus quality, short-term output versus long-term learning outcomes) can lead to divergent conclusions about whether AI “helps” or “harms.” Without explicit reporting of these dimensions, it becomes difficult to determine whether studies are estimating comparable treatment effects or answering fundamentally different questions.

Despite this heterogeneity, many papers present findings using similar high-level claims (e.g., “AI improves productivity” or “AI harms learning”). Because key contextual details are omitted or buried in appendices, results that depend on different conditions appear directly comparable when they are not. This makes it difficult to assess whether observed differences reflect true disagreement, contextual variation, or methodological artifacts. As a result, policymakers and institutional decision-makers encounter a fragmented body of causal evidence that cannot be easily synthesized. The problem is not a lack of evidence, but the absence of *standardized evidence* that allows results from human–AI collaboration experiments to accumulate into generalizable knowledge.

THE IMPLICATIONS OF MESSY STANDARDS

Human–AI research increasingly informs real-world institutional decisions. Evidence from RCTs is routinely discussed in healthcare when hospitals consider whether to give doctors AI-assisted decision-making tools, when schools debate the use of AI tutoring or writing tools, and when employers evaluate their workers’ productivity with AI tools [11, 12, 6]. In these settings, decision-makers often seek clear guidance from the scientific literature about whether AI “works”. For example, recent public missteps, such as the case of Klarna reversing its decision to replace its human customer service staff with AI-generated chatbots [13], illustrate how the general narrative that “AI helps” must be contextualized; in this example, AI helped lower costs, but it led to poorer service quality and lower customer satisfaction. By aggregating results into high-level conclusions of success or failure, the nuance of contextual variables that shape these outcomes—such as user expertise, training, and interaction design—are obscured. AI systems

may therefore be deployed in settings where they are ineffective or counterproductive, despite appearing beneficial in prior studies conducted under different conditions.

Documentation gaps also make uneven impacts difficult to detect before widespread adoption. Average effects reported in RCTs can mask heterogeneity across users or groups, particularly when prior AI familiarity, access to training, or workflow integration differ. For example, schools may adopt AI writing tools based on reported gains in average performance, only to discover later that these tools widen disparities between students with prior AI familiarity and those without; or, that the specific chat interface matters significantly—e.g., that only models fine-tuned to not give answers lead to gains [12]. Similarly, organizations may deploy AI decision-support systems without realizing gains are not universal; for instance, employees in specific operational contexts may find the tool increases friction, leading to worse performance. Without documenting the specific details of a study, we risk assuming that the result from an RCT generalizes to all settings.

Our argument is not that AI innovation should slow, or that deployment must wait for perfect understanding. Treating documentation as secondary to innovation risks erasing the very information needed to understand what works, for whom, and under what conditions. Because corrective action typically occurs only after negative outcomes become visible, the costs of misdeployment are often amplified and harder to reverse once systems are embedded in institutional practice [14]. Incomplete documentation further complicates how governments allocate funding and design interventions, by obscuring which populations and applications benefit from AI and which may be disadvantaged. Understanding when and why AI systems support or hinder human work is therefore not only a research concern, but a practical governance challenge. Without evidence that is structured to reflect how people actually interact with AI systems, institutions risk making high-stakes decisions on the basis of conclusions that do not travel across contexts.

A PATH FORWARD: REORIENTING EVALUATION

Current AI evaluation practices focus on benchmarks. Many researchers have raised concerns about the fact that models score well on benchmarks, but those scores hide weaknesses in how models might perform in other real-world settings. We emphasize calls that our understanding of AI performance must incorporate the human side: how do people actually use these AI models? It doesn't matter how an AI performs alone if, in practice, people are using the AI to make decisions. One RCT alone is unable to help us understand how humans and AI work together generally; but a body of RCT evidence might. Existing approaches demonstrate that standardized documentation can make complex evidence governable. Other research domains have dealt with this situation, where people are so busy pushing forward new findings, that after some time, people realize it is necessary to develop practices to make evidence comparable and interpretable. Three such approaches—clinical trial reporting in healthcare, preregistration in the social sciences, and model and data cards in computer science—offer complementary lessons for human–AI collaboration research.

In healthcare, RCTs are accompanied by standardized reporting frameworks that explicitly document study design, participant characteristics, interventions, and outcomes. These reporting standards ensure that results from individual trials could be evaluated, replicated, and synthesized for clinical and regulatory decision-making. For instance, clinical trials are used to assess mechanical medical devices, new methods of diagnosing and screening for disease, and novel ways to use and combine existing medications to both treat and prevent disease.

In the social sciences, transparency practices like preregistration address a different but related challenge: distinguishing robust findings from those that depend on analytic flexibility or selective reporting. Preregistration requires researchers to specify hypotheses, outcomes, and analysis plans in advance, forcing researchers to clarify what questions their study is designed to answer [15]. This simultaneously improves comparability across studies by making research decisions explicit.

In computer science, documentation frameworks such as model cards and datasheets were introduced to address the difficulty of interpreting system performance without context [16, 17]. These tools require standardized descriptions of a model's training data, intended use, and known limitations, allowing users to assess whether a system evaluated in one setting is appropriate for deployment in another. Model and data cards do not prescribe how models should be built or used; instead, they provide a shared language for communicating essential contextual information.

Taken together, these approaches highlight a common principle: when evidence is used to guide high-stakes decisions, documentation is necessary to ensure quality and safe use. Yet none of these solutions alone is sufficient for human–AI collaboration research. Clinical trial reporting focuses on interventions rather than interactive systems; these standards lack comprehensive interaction standards (and generally, health outcomes are more discrete than outcomes like productivity, learning, etc. which rely on a broader range of contextual factors) [18]. Preregistration improves transparency and requires pre-study decision-making but does not ensure post hoc comparability; specifically, it lacks the standardized constraints necessary to ensure generalizability across studies. Finally, model cards document AI systems without accounting for how people interact with them. Human–AI collaboration requires a framework that integrates these insights by documenting not only the technology and the study design, but also the human participants and the interaction context in which outcomes emerge.

AI CLINICAL TRIAL CARDS

We propose that studies of human-AI collaboration be accompanied by short (one- to two-page) summaries called **AI Clinical Trial Cards**, modeled after documentation frameworks in medicine, social science, and machine learning. These cards record essential study details so that results can be meaningfully compared across studies and over time. In doing so, they transform a collection of one-off experiments into a cumulative research infrastructure, enabling researchers, policymakers, and organizations to distinguish between genuinely conflicting findings and context-dependent results. With continued use, they also help to surface the key

dimensions along which a design space for human–AI collaboration can be built. Below we describe the priority components to address: the model used, the characteristics of participants, and the design of the interaction.

AI Model Details. This component specifies what system was used and how it was configured. This includes information such as the model provider, model name and version, fine-tuning status, and configuration parameters (e.g., temperature, prompt design, or additional constraints). These details are crucial for interpreting results: for instance, whether a study used GPT-4 or an earlier model, or whether it employed a customized versus out-of-the-box configuration, can dramatically shape outcomes.

Human Participant Profile. The second component describes who the human collaborators were. This section includes sample size, participant expertise, demographics, and any training or onboarding participants received before engaging with the AI system. A study conducted with novice undergraduate participants will differ meaningfully from one involving domain experts or professional workers. Making these differences explicit allows researchers and readers to contextualize variation in results across studies.

Interaction and Task Design. The third component outlines how humans and AI systems interacted and on what tasks. This includes the interface used (e.g., chat window, text editor, or decision-support tool), the degree of autonomy (assistive vs. semi- or fully-autonomous AI), the nature of the task, and the metrics used to evaluate success (e.g., speed, quality, or accuracy). For example, whether an AI provides real-time writing suggestions or produces outputs for later human review represents a fundamental design choice that can alter both user behavior and performance.

Together, these three elements provide a concise *TL;DR* of an experiment’s setup—capturing the essential factors that shape human–AI outcomes. Two studies might both test “AI assistance in writing tasks” yet reach opposite conclusions if one uses a weaker model or a fundamentally different workflow. The AI Clinical Trial Card allows such differences to be immediately visible, reducing the cognitive burden of inferring experimental details from dense papers. By making contextual factors transparent, it enables decision-makers to compare “apples to apples” when interpreting results across studies, rather than speculating about hidden discrepancies.

LOOKING FORWARD

The proposed AI Clinical Trial Card allows stakeholders to generate insights about how people interact with AI models—insights that go beyond standard machine learning performance benchmarks focused on isolated experimental results. By documenting contextual factors such as user expertise, task type, and interface design, these cards make it possible to understand the *human side* of AI evaluation: how people adapt, where AI succeeds, and where it fails. Crucially, this structure supports learning over time by allowing evidence to be compared, revisited, and reinterpreted as technologies and use cases evolve.

The value of such documentation extends across multiple stakeholder groups. Policymakers can draw on a transparent record of human–AI studies to inform regulation, labor policy, and governance frameworks that reflect real-world use rather than AI models’ abstract capabilities. Organizations can make more informed decisions about adopting AI technologies for specific domains (e.g. marketing or software development, based on accumulated evidence that is likely to generalize to their specific contexts). Researchers gain the ability to synthesize findings across disciplines, identify which experimental conditions have been tested, and coordinate future experiments where gaps remain. Model developers can better understand how AI systems perform in practice, guiding improvements to post-training and interaction design that align with how tools are actually used.

More broadly, **AI Clinical Trial Cards** create a shared language for documenting and comparing human–AI collaboration as it unfolds. Human behavior is inherently variable, and shaped by individual experiences, cognitive abilities, and context, making interaction effects difficult to anticipate or interpret. Behavioral and social scientists have long grappled with this challenge, and recent work on *integrative experiment design* emphasizes the importance of systematically mapping the design space of possible human–AI interactions [19]. By making experimental conditions explicit, the AI Clinical Trial Cards provide a foundation for building this design space—making it clear what we have already learned and what variables we have missed.¹

As AI systems increasingly shape human work, learning, and decision-making, governing their use requires not only better models, but better evidence about how people and AI systems interact in practice. Building this evidence infrastructure now ensures that rapid innovation does not come at the expense of interpretability, equity, or collective learning, and that societies retain the ability to learn from deployment rather than only react to its consequences.

¹ [hoping to expand on the design space point; it’s future looking]

REFERENCES

1. Organisation for Economic Co-operation and Development (OECD), Recommendation of the Council on Artificial Intelligence, OECD/LEGAL/0449 (2025).
<https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0449>
2. R. Bommasani et al., Advancing science- and evidence-based AI policy, *Science* 387, 459–461 (2025). <https://doi.org/10.1126/science.adu8449>
3. Foundations for Evidence-Based Policymaking Act of 2018, H.R. 4174, 115th Congress (2018).
<https://www.congress.gov/bill/115th-congress/house-bill/4174>
4. Office of Governor Gavin Newsom, The California Report on Frontier AI Policy (State of California, 2025).
<https://www.gov.ca.gov/wp-content/uploads/2025/06/June-17-2025-%E2%80%93-The-California-Report-on-Frontier-AI-Policy.pdf>
5. E. Brynjolfsson, D. Li, and L. Raymond, Generative AI at work, *Quarterly Journal of Economics* (2025).
6. F. Dell'Acqua et al., Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality, Harvard Business School Working Paper No. 24-013 (2023).
https://www.hbs.edu/ris/Publication%20Files/24-013_d9b45b68-9e74-42d6-a1c6-c72fb70c7282.pdf
7. S. Noy and W. Zhang, Experimental evidence on the productivity effects of generative artificial intelligence, *Science* (2023). <https://doi.org/10.1126/science.adh2586>
8. N. Otis, R. Clarke, D. Holtz, and R. Koning, Uneven impacts of generative AI on entrepreneurial performance, Working Paper (2024).
9. A. Humlum and E. Vestergaard, Large language models, small labor market effects, National Bureau of Economic Research Working Paper No. 32404 (2025).
10. National Institute of Standards and Technology (NIST), Artificial Intelligence Risk Management Framework (AI RMF 1.0) (NIST, 2023). <https://www.nist.gov/itl/ai-risk-management-framework>
11. E. Goh et al., GPT-4 assistance for improvement of physician performance on patient care tasks: A randomized controlled trial, *Nature Medicine* 30 (2025).
<https://www.nature.com/articles/s41591-024-03456-y>
12. H. Bastani et al., Generative AI without guardrails can harm learning: Evidence from high school mathematics, *Proceedings of the National Academy of Sciences* (2025).
<https://doi.org/10.1073/pnas.2422633122>
13. M. Gene, "Business Tech News: Klarna Reverses On AI, Says Customers Like Talking To People." *Forbes*. (2025).
14. D. Collingridge, *The social control of technology*. Frances Pinter (1982).
15. B. A. Nosek et al., "The preregistration revolution." *Proceedings of the National Academy of Sciences* 115.11 (2018): 2600-2606.
16. T. Gebru et al., Datasheets for datasets. *Communications of the ACM*, 64(12), 86–92 (2021)
17. M. Mitchell et al., Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 220–229). ACM (2019).
18. I. D. Liu et al., CONSORT-AI extension: Reporting guidelines for clinical trials evaluating artificial intelligence interventions, *Nature Medicine* (2020).
<https://www.nature.com/articles/s41591-020-1034-x>
19. A. Almaatouq et al., "Beyond Playing 20 Questions with Nature: Integrative Experiment Design in the Social and Behavioral Sciences." *Behavioral and Brain Sciences* 47 (2024).

APPENDIX

AI Clinical Trial Card
[Design and specifics in progress]

AI: What model was used, and how was it configured?

- Provider: e.g., OpenAI, Google, Anthropic, Meta
- Model Name & Version: e.g., gpt-4-turbo-2024-04-09, claude-3-opus-20240229
- Access Method: e.g., API, Web Interface (ChatGPT Plus), Local Instance
- Key Parameters: e.g., Temperature, Top-P, System Prompt (full text if used)
- Performance on Standard Benchmarks: e.g., MMLU, GSM8K, HumanEval scores for the specific model version used, if available.
- Situates the model's general capabilities at the time of the experiment.

Human: Who were the participants, and what expertise did they bring?

- Data Collection Period: Start Date – End Date
- # of participants (N)
- Domain expertise
- LLM familiarity

Interaction/Task: What did they do together, and how was success measured?

- Task: Short description (e.g., “consultants solving business cases”)
- Conditions: Human-only / AI-only / Human+AI
- Interface: Describe the interface through which the user interacted with the model.
- Participant Prompting Strategy: e.g., Zero-shot, Few-shot, Chain-of-Thought. Was any training or tutorial on prompting provided?
- Outcome Metric(s): Metric and how measured
- % improvement

Figure 1: A set of possible information to put in the card.